

Advanced Econometrics

1 General-to-Specific Approach

Exercise 1

For data in `nlsw88.dta` file apply general-to-specific procedure to the model:

$$\begin{aligned} wage = & \beta_0 + \beta_1 age + \beta_2 race + \beta_3 married + \beta_4 grade + \beta_5 south + \beta_6 union + \\ & + \beta_7 hours + \beta_8 ttl_exp + \beta_9 tenure + \beta' industry + \varepsilon \end{aligned}$$

This dataset comes from the textbook by Christopher F. Baum *An Introduction to Modern Econometrics Using Stata*. It is an extract of the U.S. National Longitudinal Survey (NLSW) for employed women in 1988. We restrict the sample of 2,246 women to a subsample for which data on an indicator of *union* is available. This step reduces the sample to 1,865 women.

Exercise 2

Import data from `Artificial_data.txt`. Apply general-to-specific approach for variable selection in the regression model for variable *y*. Present all the estimated models in one publication quality table.

Exercise 3

What is the relationship between crime and punishment? This important question has been examined by Cornwell and Trumbull¹ using a panel of data from North Carolina. The cross sections are 90 counties, and the data are annual for the years 1981–1987. The data are in the file `crime.csv`.²

The code in Exercise 3 section is the first few steps of the general-to-specific procedure. Try to complete the procedure.

Variables description

1. <code>county</code>	county identifier
2. <code>year</code>	81 to 87
3. <code>crmte</code>	crimes committed per person
4. <code>prbarr</code>	'probability' of arrest
5. <code>prbconv</code>	'probability' of conviction
6. <code>prbpris</code>	'probability' of prison sentenc
7. <code>avgsen</code>	avg. sentence, days
8. <code>polpc</code>	police per capita
9. <code>density</code>	people per sq. mile
10. <code>taxpc</code>	tax revenue per capita
11. <code>west</code>	=1 if in western N.C.
12. <code>central</code>	=1 if in central N.C.

¹"Estimating the Economic Model of Crime with Panel Data," *Review of Economics and Statistics*, 76, 1994, 360–366.

²R. Carter Hill, William E. Griffiths and Guay C. Lim, *Principles of Econometrics*, 4th Edition, p.124

13. urban	=1 if in SMSA
14. pctmin80	perc. minority, 1980
15. wcon	weekly wage, construction
16. wtuc	wkly wge, trns, util, commun
17. wtrd	wkly wge, whlesle, retail trade
18. wfir	wkly wge, fin, ins, real est
19. wser	wkly wge, service industry
20. wmfg	wkly wge, manufacturing
21. wfed	wkly wge, fed employees
22. wsta	wkly wge, state employees
23. wloc	wkly wge, local gov emps
24. mix	offense mix: face-to-face/other
25. pctymle	percent young male
26. d82	=1 if year == 82
27. d83	=1 if year == 83
28. d84	=1 if year == 84
29. d85	=1 if year == 85
30. d86	=1 if year == 86
31. d87	=1 if year == 87
32. lcrmrte	log(crmrte)
33. lprbarr	log(prbarr)
34. lprbconv	log(prbconv)
35. lprbpris	log(prbpris)
36. lavgsen	log(avgsen)
37. lpolpc	log(polpc)
38. ldensity	log(density)
39. ltaxpc	log(taxpc)
40. lwcon	log(wcon)
41. lwtuc	log(wtuc)
42. lwtrd	log(wtrd)
43. lwfir	log(wfir)
44. lwser	log(wser)
45. lwmfg	log(wmfg)
46. lwfed	log(wfed)
47. lwsta	log(wsta)
48. lwloc	log(wloc)
49. lmix	log(mix)
50. lpctymle	log(pctymle)
51. lpctmin	log(pctmin)
52. clcrmrte	lcrmrte - lcrmrte[t-1]
53. clprbarr	lprbarr - lprbarr[t-1]
54. clprbconv	lprbconv - lprbconv[t-1]
55. clprbpri	lprbpri - lprbpri[t-1]
56. clavgsen	lavgsen - lavgsen[t-1]
57. clpolpc	lpolpc - lpolpc[t-1]
58. cltaxpc	ltaxpc - ltaxpc[t-1]
59. clmix	lmix - lmix[t-1]